Car Price Analysis Report

Comprehensive analysis of 2,500 cars to understand pricing trends across brands, models, and market segments.

Project Overview

Dataset

2,500 records with brand, model, price, year, fuel type, transmission, and condition features.

Data Cleaning

Python and Pandas for preprocessing, handling missing values, and creating price categories.

SQL Analysis

Query-based analysis for brand comparisons, pricing trends, and premium segment identification.

Visualization

Power BI dashboard for interactive exploration of pricing insights and business intelligence.

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Dataset Summary

Key Characteristics

- 2,500 records with comprehensive car attributes
- Well-structured and evenly distributed across brands
- Features include brand, model, price, year, fuel type, transmission, and condition
- Even brand distribution ensures unbiased analysis

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SQL Analysis: Overview & Distribution

2,500

Total Cars

Complete dataset size

$52,638

Average Price

Mid-range market positioning

7

Brands

Evenly distributed

Brand Distribution

Brand	Car Count
Audi	368
BMW	358
Tesla	348
Honda	352
Ford	347
Mercedes	365
Toyota	362

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SQL Analysis: Brand Pricing Comparison

Average Price by Brand

Brand	Average Price
BMW	$54,157
Tesla	$53,476
Mercedes	$53,191
Honda	$52,050
Toyota	$52,079
Audi	$51,953
Ford	$51,593

Insight: BMW and Tesla emerge as premium brands, while Ford offers relatively affordable options.

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SQL Analysis: Premium Segment

Top 5 Most Expensive Cars

Brand	Model	Price
Audi	A3	$99,983
BMW	3 Series	$99,969
Tesla	Model S	$99,906
Tesla	Model 3	$99,794
Mercedes	E-Class	$99,754

Insight: Most expensive cars cluster near $100K, showing premium segment concentration.

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SQL Analysis: Additional Insights

Year-wise Trends

Prices fluctuate but remain stable overall, with no significant upward or downward trend.

Top 3 Models per Brand

Each brand offers multiple high-value models, providing diverse premium options.

Above Average Cars

BMW leads with 200 cars above brand average, followed by Toyota (187) and Honda (179).

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Power BI Dashboard

Key Visualizations

- Total and average price by brand (bar chart)
- Average price by condition (horizontal bar chart)
- Average price by model (table)
- Average price by fuel type (donut chart)
- Average price by transmission (pie chart)

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Dashboard KPIs

2.5K

Total Cars

$52.64K

Avg Price

$131.6M

Total Value

3.47

Avg Engine Size

149.75K

Avg Distance Driven

Fuel Type Distribution

- Hybrid: 25.97%
- Diesel: 24.9%
- Electric: 24.97%
- Petrol: 24.58%

Transmission Distribution

- Automatic: 68%
- Manual: 32%

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Business Recommendations

Focus Marketing

Target premium brands like BMW and Tesla for high-value customer segments.

Optimize Pricing

Refine pricing strategy for mid-range cars to maximize market penetration.

Analyze Models

Study high-value models for better positioning and competitive advantage.

Customer Segmentation

Use insights for targeted segmentation and personalized marketing campaigns.